

# Assignment 3

Diya Jain  
V81673988

1. Alice transmitted the string: 1101 0111 0010 0011 0011  
Using the following measurement bases: 11-1 ---1 -1-1 -11- 1--1  
Bob ~~transmitted the string~~ used the measurement bases:  
1-1- 111- -1-1 11-1 1--1

Using the above things, bob should interpret the string as: 1RRR RRRR 0010 R0RR 0011, where R = random string.

But bob recieved the following string: 1010 1000 1111 1100 0001

We can see that there is not a 100% str match between the strings. This shows that the message has been tampered with or someone has been eavesdropping.

$$2. \begin{bmatrix} -2/\sqrt{3} \\ -\sqrt{3}/3 \\ 2\sqrt{8}/\sqrt{7} \\ \sqrt{24}/63 \end{bmatrix} = \begin{bmatrix} A_1 \times B_1 \\ A_1 \times B_2 \\ A_2 \times B_1 \\ A_2 \times B_2 \end{bmatrix} = A_1 \otimes B_1 = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \otimes \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

$$\text{So, } \frac{1}{3\sqrt{7}} \begin{bmatrix} -2 \\ -\sqrt{3} \\ 2\sqrt{8} \\ \sqrt{24} \end{bmatrix} = \begin{bmatrix} -1 \times 2 \\ -1 \times \sqrt{3} \\ \sqrt{8} \times 2 \\ \sqrt{8} \times \sqrt{3} \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ \sqrt{8} \end{bmatrix} \otimes \begin{bmatrix} 2 \\ \sqrt{3} \end{bmatrix}$$

↑                    ↑  
A                    B

$$3. a. (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$$

$$\left( \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \otimes \left( \gamma \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \delta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$\left( \begin{bmatrix} \alpha \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \beta \end{bmatrix} \right) \otimes \left( \begin{bmatrix} \gamma \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \delta \end{bmatrix} \right)$$

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix}$$

b. We are given 2 orthonormal bases

$$B_1 = \{|\psi_1\rangle, |\psi_2\rangle\} \text{ and } B_2 = \{|\phi_1\rangle, |\phi_2\rangle\}$$

So, we can say that

$$\langle \psi_i | \psi_j \rangle = \begin{cases} 1 & \text{when } i=j \\ 0 & \text{when } i \neq j \end{cases} \quad \text{and}$$

$$\langle \phi_m | \phi_n \rangle = \begin{cases} 1 & \text{when } m=n \\ 0 & \text{when } m \neq n \end{cases}$$

To prove  $B_3 = \{|\psi_1\rangle \otimes |\phi_1\rangle, |\psi_1\rangle \otimes |\phi_2\rangle, |\psi_2\rangle \otimes |\phi_1\rangle, |\psi_2\rangle \otimes |\phi_2\rangle\}$   
we can say that  $\langle v | w \rangle = (\langle \psi_i | \otimes \langle \phi_m |)(|\psi_j\rangle \otimes |\phi_n\rangle)$

$$\{ \langle v | w \rangle = \langle \psi_i | \psi_j \rangle \otimes \langle \phi_m | \phi_n \rangle$$

$$= \langle \psi_i | \psi_j \rangle \cdot \langle \phi_m | \phi_n \rangle = \begin{cases} 1 & \text{when } i=j, m=n \\ 0 & \text{when } i \neq j, m \neq n \end{cases}$$

that means that  $\langle v | w \rangle = 0$  for all distinct  $|v\rangle, |w\rangle \in B_3$  and  $\|v\|^2 = 1$

$\therefore$  this proves that all 4 elements of the set  $B_3$  are orthonormal.